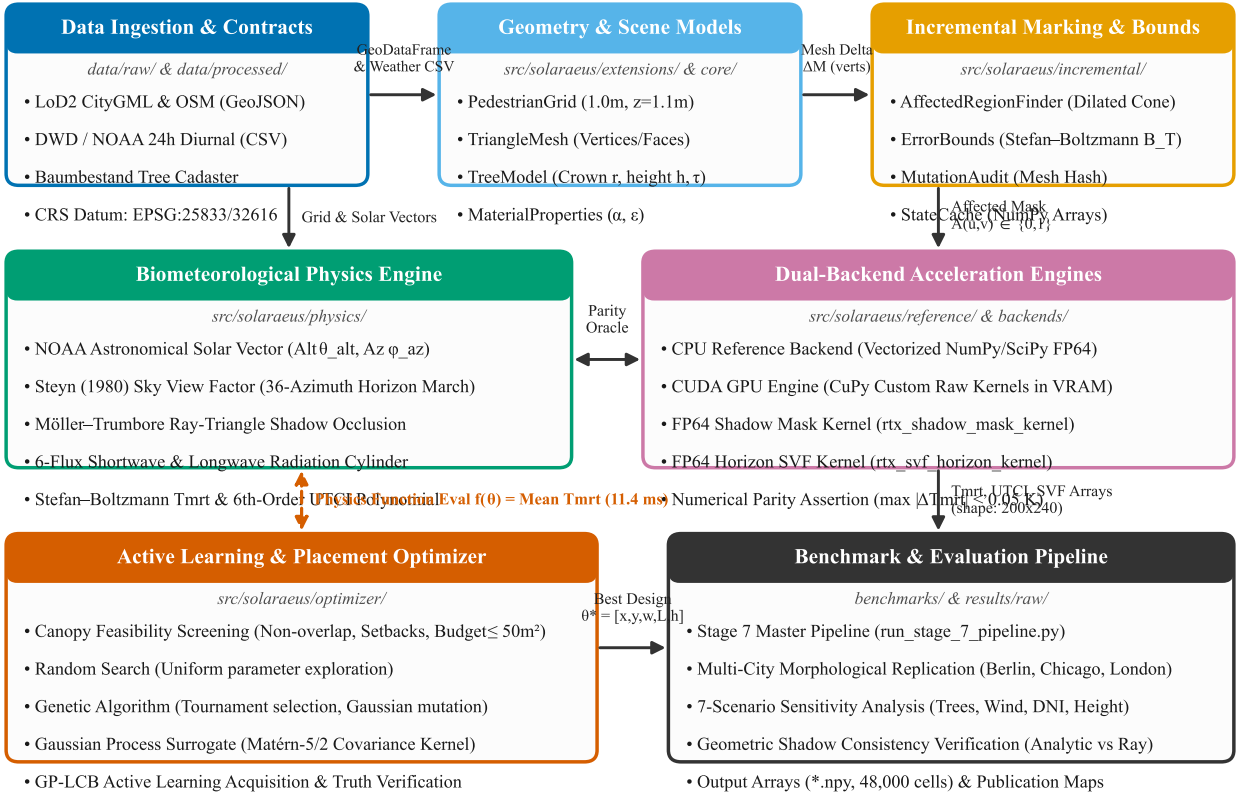

Data Ingestion & Contracts
data/raw/ & data/processed/
• LoD2 CityGML & OSM (GeoJSON)
• DWD / NOAA 24h Diurnal (CSV)
• Baumbestand Tree Cadaster
• CRS Datum: EPSG:25833/32616
GeoDataFrame & Weather CSV
Geometry & Scene Models
src/solaraeus/extensions/ & core/
• PedestrianGrid (1.0m, z=1.1m)
• TriangleMesh (Vertices/Faces)
• TreeModel (Crown r, height h, τ)
• MaterialProperties (α, ε)
Mesh Delta ΔM (verts)
Incremental Marking & Bounds
src/solaraeus/incremental/
• AffectedRegionFinder (Dilated Cone)
• ErrorBounds (Stefan–Boltzmann B_T)
• MutationAudit (Mesh Hash)
• StateCache (NumPy Arrays)
Grid & Solar Vectors
Affected Mask A(u,v) ∈ {0,1}
Biometeorological Physics Engine
src/solaraeus/physics/
• NOAA Astronomical Solar Vector (Alt θ_alt, Az φ_az)
• Steyn (1980) Sky View Factor (36-Azimuth Horizon March)
• Möller–Trumbore Ray-Triangle Shadow Occlusion
• 6-Flux Shortwave & Longwave Radiation Cylinder
• Stefan–Boltzmann Tmrt & 6th-Order UTCI Polynomial
Parity Oracle
Dual-Backend Acceleration Engines
src/solaraeus/reference/ & backends/
• CPU Reference Backend (Vectorized NumPy/SciPy FP64)
• CUDA GPU Engine (CuPy Custom Raw Kernels in VRAM)
• FP64 Shadow Mask Kernel (rtx_shadow_mask_kernel)
• FP64 Horizon SVF Kernel (rtx_svf_horizon_kernel)
• Numerical Parity Assertion (max |ΔTmrt| < 0.05 K)
UTCI Polynomial Eval f(θ) = Mean Tmrt (11.4 ms)
Tmrt, UTCI, SVF Arrays (Shape: 200X240)
Active Learning & Placement Optimizer
src/solaraeus/optimizer/
• Canopy Feasibility Screening (Non-overlap, Setbacks, Budget≤ 50m²)
• Random Search (Uniform parameter exploration)
• Genetic Algorithm (Tournament selection, Gaussian mutation)
• Gaussian Process Surrogate (Matérn-5/2 Covariance Kernel)
• GP-LCB Active Learning Acquisition & Truth Verification
Best Design θ* = [x,y,w,L,h]
Benchmark & Evaluation Pipeline
benchmarks/ & results/raw/
• Stage 7 Master Pipeline (run_stage_7_pipeline.py)
• Multi-City Morphological Replication (Berlin, Chicago, London)
• 7-Scenario Sensitivity Analysis (Trees, Wind, DNI, Height)
• Geometric Shadow Consistency Verification (Analytic vs Ray)
• Output Arrays (*.npy, 48,000 cells) & Publication Maps